SHoEbox: residual byte decoders

2026-11-04 | Frozen linear path, 16-unit corrections, validation-only selection

The residual preserves most baseline behavior but does not solve the shared/private tradeoff. Validation chooses zero strength for 22 of 36 GELU heads and 33 of 36 matched identity heads. Selected GELU improves B-only recovery on dropout-trained graphs, while private-answer accuracy remains nearly unchanged. The default decoder stays unchanged.

Private answers on held-out combinations

Frozen graph	Frozen linear	Identity selected	GELU full	GELU selected
Original / dropout	73.7%	72.9%	72.6%	73.7%
Original / balanced	56.0%	56.0%	61.2%	56.0%
Gated / dropout	60.9%	60.9%	62.5%	60.9%
Gated / balanced	54.9%	54.9%	62.4%	55.2%

Shared answers on held-out combinations

Frozen graph	Frozen linear	Identity selected	GELU full	GELU selected
Original / dropout	87.0%	87.5%	82.0%	87.2%
Original / balanced	80.5%	80.5%	66.3%	80.5%
Gated / dropout	91.1%	91.1%	85.3%	90.5%
Gated / balanced	87.5%	87.5%	74.7%	87.4%

At full strength, the balanced-source GELU corrections improve private accuracy from **56.0% to 61.2%** under original publication and **54.9% to 62.4%** under evidence gating. Shared accuracy falls by **14.1 and 12.8 percentage points**. Validation therefore backs off most corrections, also removing their private-answer gains.

Overall held-out accuracy

Frozen graph	Frozen linear	Identity selected	GELU selected
Original / dropout	78.1%	77.8%	78.2%
Original / balanced	64.1%	64.1%	64.1%
Gated / dropout	71.0%	71.0%	70.8%
Gated / balanced	65.8%	65.8%	65.9%

Each row averages three frozen graph seeds. New decoder results first average three initialization seeds within each graph. Every head emits one of 256 bytes from the same observed raw feature vector; query labels never select a decoder or enter its prediction interface. Full-strength results are diagnostics, not deployment selections.

B-only shared recovery

Frozen graph	Frozen linear	Prior GELU	Selected residual GELU
Original / dropout	85.7%	97.6%	90.1%
Original / balanced	100.0%	99.8%	100.0%
Gated / dropout	95.8%	99.1%	97.3%
Gated / balanced	98.7%	99.9%	98.6%

B-only evaluation removes A and C and covers all 64 fact combinations. The all-source held-out task above uses 16 combinations excluded from fitting. These conditions share the same frozen models but test different information paths.

What validation selected

Strength	Identity heads	GELU heads
0	33	22
0.125	0	10
0.25	3	4
0.5	0	0
1	0	0

One scalar per head is chosen from the fixed grid **0, 0.125, 0.25, 0.5, 1**. Selection minimizes query-balanced validation cross entropy, subject to both all-source/shared and B-only/shared exact-correct counts being at least their frozen-linear baseline. Ties within 1e-12 prefer the smaller strength. Zero is always feasible and reproduces the baseline exactly.

For 14 of the 22 zero-selected GELU heads, every nonzero candidate violates a shared-answer guard. For the other eight, an eligible nonzero candidate exists but does not improve the selection objective enough to beat zero. Identity has 15 guard-only rejections and 18 objective-based zero selections.

The guards apply to validation. Test shared accuracy changes from **91.1% to 90.5%** on gated/dropout graphs and **87.5% to 87.4%** on gated/balanced graphs. They do not guarantee preservation on unseen examples.

Variation across graphs and initializations

Frozen graph	Selected GELU private: graph range	Largest within-graph initialization range
Original / dropout	68.1% to 79.3%	2.7 pp
Original / balanced	49.2% to 62.1%	0.0 pp
Gated / dropout	55.1% to 68.4%	2.0 pp
Gated / balanced	37.5% to 69.8%	2.0 pp

Ranges describe observed seed variation, not confidence intervals. All three initializations are reported without choosing a winner. Selected GELU averages **25.1%** on missing-information controls, against a 25% balanced-answer reference rate.

Implementation and protocol

The decoder computes frozen linear logits plus a scaled correction. The correction uses the frozen head's training normalization, a 64-to-16 hidden projection, identity or GELU activation, and a 16-to-256 output projection. Its output weights and bias start at zero, giving exactly the original logits at initialization. Only correction parameters receive gradients.

Setting	Value
Frozen graphs	12: two publication policies x two source schedules x three graph seeds
Graph seeds / correction seeds	20261106-20261108 / 20261121-20261123
Fitted corrections	72: identity and GELU, three initializations, twelve graphs
Parameters per decoder	5,392 trainable correction + 16,640 frozen linear = 22,032 weights/biases
Training per graph	960 supported source/query cases; 192 distinct episode identities
Validation per graph	1,920 cases; 384 distinct identities; 16 separate fact combinations
Evaluation per graph	384 all-source held-out cases; 3,456 independent source controls
Optimizer	Full-batch L-BFGS, strong Wolfe; 400-iteration cap; L2 0.001 on correction weights/biases
Loss weights	Equal total mass per query; uniform supported subsets within query
Additional graph work	4,032,000 expert calls per graph; 48,384,000 total; K=1

Training, test and independent features are reused from the previous study. Only validation requires new graph execution. Each identity/GELU pair shares initial tensors, parameter counts and fitting settings. The prior standalone GELU comparison used 64 hidden units and 20,800 trainable parameters; this study's residual correction uses 16 units. Lower trainable count does not imply lower total inference cost because the frozen linear path remains active.

All **72 of 72 fits** reach the 400-iteration cap; the largest final coefficient gradient is **0.001440**. The fit budget is fixed across conditions. A selection-lock file records validation candidate scores and the chosen strength before evaluation runs.

Verification and reproduction

All **284 automated tests pass**, including 15 new residual checks. The live-output example also passes. Independent artifact audits reconstruct every synthetic cohort, verify frozen graph/base identities and source hashes, reproduce all saved predictions and losses, reconstruct training objectives, and reproduce every validation decision. Prior raw-linear predictions match exactly. Prediction exports are written atomically and verified before a run is marked complete.

```
python scripts/run_residual_study.py \
  --out artifacts/residual_reproduction --jobs 3 --threads 2
python -m experiments.text_regions.residual_compare \
  --source artifacts/residual_adapter_v1 \
  --out artifacts/reproduced_residual_comparison.json
python scripts/check_all.py
```

The complete download includes source, reports, original checkpoints, earlier feature caches/heads, new corrections, validation caches, selection locks, and full synthetic prediction rows. The public repository contains code, reports and aggregate results.

Next development step

Move shared-answer preservation into correction training, so fitting searches for private gains that coexist with shared accuracy. Retain the frozen baseline, parameter-matched identity control and validation-only strength selection. Post-fit scaling alone has now shown its limit: it mostly restores the old answer boundary instead of finding a better shared/private boundary.